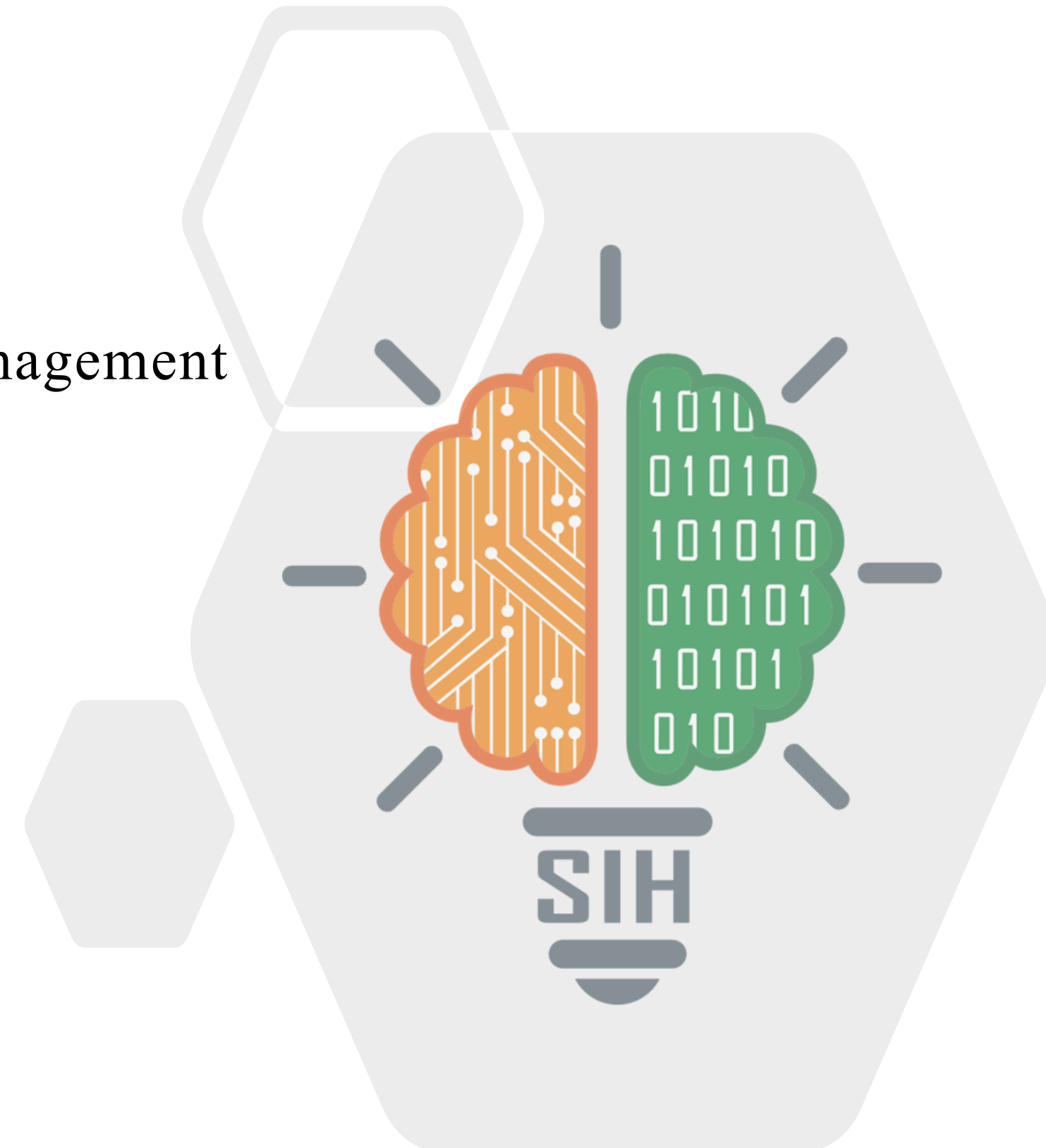


TITLE PAGE

- **Problem Statement ID** – 26190
- **Problem Statement Title-** Secure Digital Document Management System for Legal and Investigation Documents
- **Theme-**Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID** -
- **Team Name:** Innovortex



Challenges Faced: Managing vast amounts of sensitive legal documents using fragmented paper systems causes severe inefficiencies. Organizations struggle with locating files quickly while facing high risks of unauthorized access and document tampering. Poor version control and inefficient collaboration between departments further delay critical investigative processes. Ultimately the lack of proper audit tracking makes it extremely difficult to maintain data integrity and compliance as case volumes grow.



Proposed Solution:

Unified Platform - Web based application for police, lawyers, forensic experts and judges

Centralized Dashboard - Use of only single dashboard for all the occupations

Optimal Security -Ensures Authentication security especially for uploading verified reports and documents by an encrypted file vault

Transparency - Ensures smooth collaboration between officials

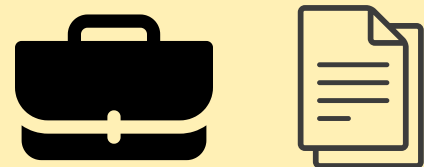
Uniqueness of Solution:

- Secure Role-Based Login
- Verified Forensic Reports
- Direct e-FIR Integration
- Need-to-Know Access
- Automatic Case Locking

EXISTING CHALLENGES

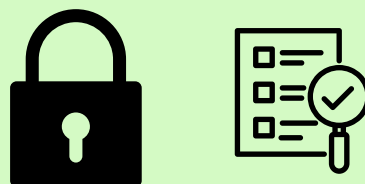
Fragmented & Paper-Based Files
Unverifiable Chain of Custody
Siloed Inter-Agency Workflows
Post-Verdict Tampering Risks

DASTAVULT



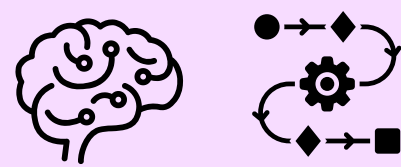
Case And Document
Management

e-FIR Auto-Linking
Centralized Case Vaults
Off-Chain S3 Storage
Document Versioning



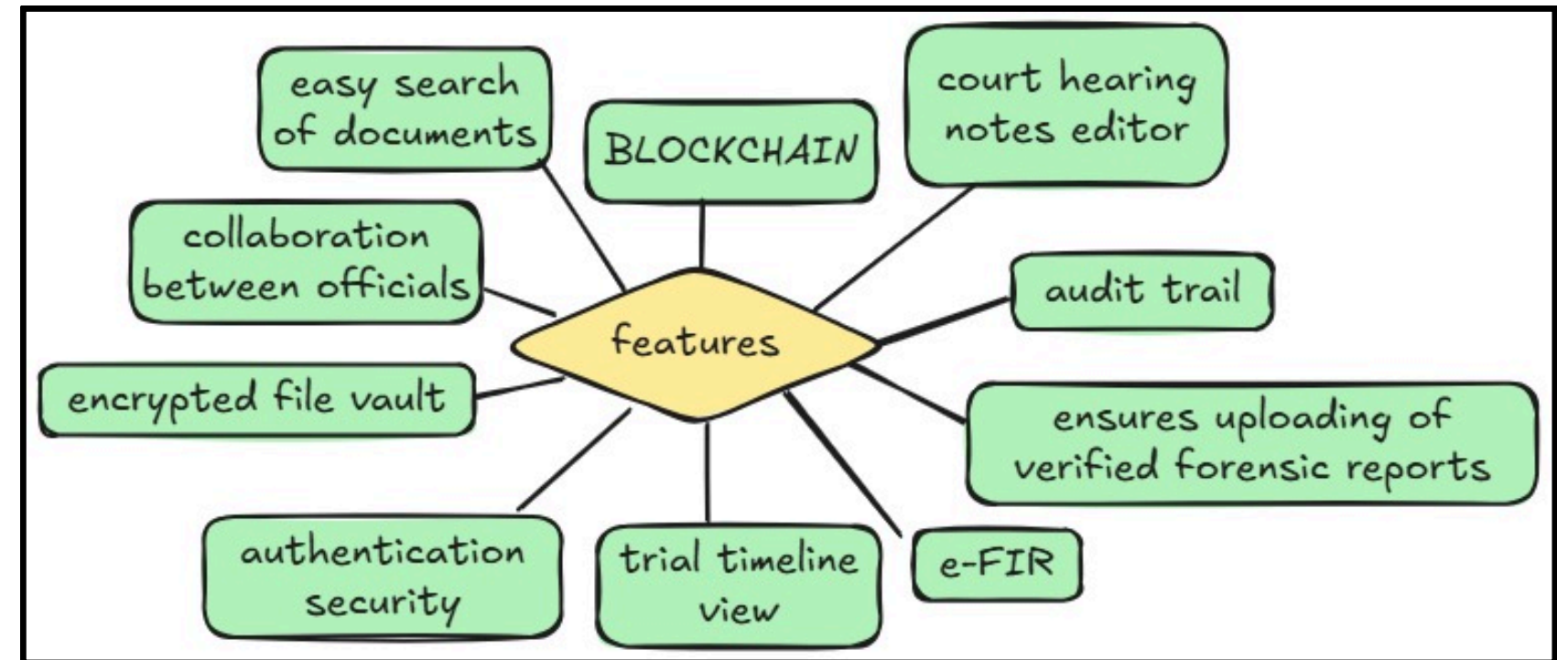
High Security

Hyperledger Audit Logs
OTP + Badge Auth
Post-Verdict Auto-Lock
AES-256 Encryption



Smart Workflow and
Integration

Role-Based Dashboards
Real-Time Case Tracking
Cross-Agency Access
BSA 2023 Compliant



TECH STACK:

- Database: MongoDB and PostgreSQL
- Backend: Node.js and Express – For robust and scalable server logic.
- Frontend: React, Tailwind CSS — For responsive dashboard interfaces.
- Blockchain: Hyperledger Fabric, Smart Contracts, IPFS — For immutable audit logs.
- Authentication: Firebase — For secure token and user management
- Security & Vault: AES-256 Encryption — For encrypted case storage.
- Integrity Layer: SHA-256 Hashing — For tamper-proof verification.



Technical Feasibility
Integrates Hyperledger blockchain with existing e-FIR APIs for a secure, government-ready architecture



Operational Feasibility
Mirrors existing legal workflows with zero-trust, admin-gated access to ensure high adoption.



Economic Feasibility
Eliminates massive physical storage overhead through highly scalable, usage-based cloud infrastructure.



Legal and Regulatory Feasibility
Ensures evidence admissibility under BSA 2023 (via blockchain) and DPDP compliance

	Paper-based / fragmented digital storage		Centralized, digitized, cloud-based document vault
	Unauthorized access to sensitive case files		Role-based access control + OTP + Admin verification
	Document tampering risk		Blockchain-hashed, immutable audit trail for every action
	No version control / edit history		Full activity logging with timestamped blockchain records
	Poor collaboration across departments		Unified platform connecting police, lawyers, forensics, judges
	Delays due to manual FIR/document handling		Direct e-FIR portal integration, instant case linking
	Weak auditability & compliance tracking		Blockchain audit trail + Admin dashboard for oversight
	No safeguard after case closure		Auto-lock of case folder post-final-verdict

DEMO LINK



Police Officer

1. Faster access to FIRs and case history.
2. Reduced paperwork
3. Better coordination with other departments



Judges and Lawyers

1. Reduces dependency on physical files
2. Helps in faster decision-making
3. Improves transparency



Forensic Agency

1. Prevents evidence tampering
2. Faster sharing of forensic results
3. Maintains evidence history



Admin

1. Centralized control over the system
2. Transparency through audit logs
3. User access management
4. Department-level monitoring



Environmental Benefits

1. Digital Preservation
2. Reduction in Paper Usage



Social Benefits

1. Faster Justice Delivery
2. Better Coordination Between Authorities



Economic Benefits

1. Improved Productivity
2. Efficient Resource Management

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